

Lagrangian Mechanics

Lecture Notes

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1 Introduction

Lagrangian mechanics is an energy-based formulation of classical mechanics. It is especially useful for systems with constraints and generalized coordinates.

2 Generalized Coordinates

Instead of Cartesian coordinates, we use generalized coordinates:

$$q_1, q_2, \dots, q_n$$

3 Degrees of Freedom

The number of independent coordinates required to describe a system.

4 The Lagrangian

$$L = T - V$$

where T is kinetic energy and V is potential energy.

5 Euler–Lagrange Equation

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

6 Procedure

1. Choose generalized coordinates
2. Write T and V
3. Form $L = T - V$
4. Apply Euler-Lagrange equation

7 Solved Problems

7.1 Problem 1: Free Particle

$$L = \frac{1}{2}m\dot{x}^2$$
$$\frac{d}{dt}(m\dot{x}) = 0 \Rightarrow \ddot{x} = 0$$

7.2 Problem 2: Harmonic Oscillator

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$$
$$m\ddot{x} + kx = 0$$

7.3 Problem 3: Simple Pendulum

$$L = \frac{1}{2}ml^2\dot{\theta}^2 - mgl(1 - \cos \theta)$$
$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0$$

7.4 Problem 4: Particle in Gravity

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy$$
$$\ddot{x} = 0, \quad \ddot{y} = -g$$

7.5 Problem 5: Polar Coordinates

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r)$$
$$m\ddot{r} - mr\dot{\theta}^2 = -\frac{dV}{dr}$$

7.6 Problem 6: Bead on Wire

$$T = \frac{1}{2}m(1 + (f'(x))^2)\dot{x}^2, \quad V = mgf(x)$$

7.7 Problem 7: Two-Mass System

$$L = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 - \frac{1}{2}k(x_1 - x_2)^2$$

7.8 Problem 8: Central Force

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r)$$
$$mr^2\dot{\theta} = \text{constant}$$

7.9 Problem 9: Charged Particle

$$L = \frac{1}{2}m\dot{x}^2 - qEx$$
$$m\ddot{x} = qE$$

7.10 Problem 10: Rolling Cylinder

$$T = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$
$$T = \frac{3}{4}mv^2$$

8 Key Takeaways

- Lagrangian mechanics is based on energy
- Works well with constraints
- Leads to conservation laws

Lagrangian mechanics is an energy-based formulation of classical mechanics. It is especially useful for systems with constraints and generalized coordinates.

9 Generalized Coordinates

We describe a system using generalized coordinates:

$$q_1, q_2, \dots, q_n$$

10 Degrees of Freedom

The number of independent coordinates required to describe the system.

11 Constraints

- Holonomic: $f(q_1, q_2, t) = 0$
- Non-holonomic: velocity dependent

12 Lagrangian

$$L = T - V$$

13 Euler–Lagrange Equation

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

14 Procedure

1. Choose generalized coordinates
2. Compute kinetic energy T
3. Compute potential energy V
4. Form $L = T - V$
5. Apply Euler–Lagrange equation

15 Solved Problems

15.1 Problem 1: Free Particle

$$L = \frac{1}{2}m\dot{x}^2$$

$$\frac{d}{dt}(m\dot{x}) = 0 \Rightarrow \ddot{x} = 0$$

Solution:

$$x = vt + c$$

15.2 Problem 2: Simple Harmonic Oscillator

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$$

$$m\ddot{x} + kx = 0$$

$$x = A \cos(\omega t + \phi), \quad \omega = \sqrt{\frac{k}{m}}$$

15.3 Problem 3: Simple Pendulum

$$L = \frac{1}{2}ml^2\dot{\theta}^2 - mgl(1 - \cos \theta)$$

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0$$

Small angle:

$$\ddot{\theta} + \frac{g}{l} \theta = 0$$

15.4 Problem 4: Particle in Gravity

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy$$

$$\ddot{x} = 0, \quad \ddot{y} = -g$$

15.5 Problem 5: Polar Coordinates

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r)$$

$$m\ddot{r} - mr\dot{\theta}^2 = -\frac{dV}{dr}$$

$$\frac{d}{dt}(mr^2\dot{\theta}) = 0$$

15.6 Problem 6: Atwood's Machine

$$T = \frac{1}{2}(m_1 + m_2)\dot{x}^2$$

$$V = (m_1 - m_2)gx$$

$$L = \frac{1}{2}(m_1 + m_2)\dot{x}^2 - (m_1 - m_2)gx$$

$$\ddot{x} = \frac{(m_2 - m_1)g}{m_1 + m_2}$$

15.7 Problem 7: Bead on a Wire

$$T = \frac{1}{2}m(1 + (f'(x))^2)\dot{x}^2$$

$$V = mgf(x)$$

$$L = \frac{1}{2}m(1 + f'^2)\dot{x}^2 - mgf(x)$$

15.8 Problem 8: Double Mass System

$$L = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 - \frac{1}{2}k(x_1 - x_2)^2$$
$$m\ddot{x}_1 = -k(x_1 - x_2)$$
$$m\ddot{x}_2 = k(x_1 - x_2)$$

15.9 Problem 9: Charged Particle

$$L = \frac{1}{2}m\dot{x}^2 - qEx$$
$$m\ddot{x} = qE$$

15.10 Problem 10: Rolling Cylinder

$$T = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

For solid cylinder $I = \frac{1}{2}mR^2$:

$$T = \frac{3}{4}mv^2$$
$$a = \frac{2}{3}g \sin \theta$$

16 Important Concepts

16.1 Cyclic Coordinates

If:

$$\frac{\partial L}{\partial q} = 0$$

then momentum is conserved.

16.2 Conservation Laws

- Time invariance $\rightarrow$ Energy conserved
- Space invariance $\rightarrow$ Linear momentum conserved
- Rotational invariance $\rightarrow$ Angular momentum conserved

17 Conclusion

Lagrangian mechanics provides a powerful framework for analyzing mechanical systems, especially those with constraints and non-Cartesian coordinates. Lagrangian mechanics provides a powerful formulation of classical mechanics based on energy methods. It is especially useful for systems with constraints and non-Cartesian coordinates.

18 Generalized Coordinates

18.1 Definition

A system is described using generalized coordinates:

$$q_1, q_2, \dots, q_n$$

which uniquely define the configuration of the system.

18.2 Relation to Cartesian Coordinates

$$x_i = x_i(q_1, q_2, \dots, q_n, t)$$
$$\dot{x}_i = \sum_j \frac{\partial x_i}{\partial q_j} \dot{q}_j + \frac{\partial x_i}{\partial t}$$

18.3 Degrees of Freedom

The number of independent generalized coordinates equals the degrees of freedom of the system.

19 Generalized Velocity

$$\dot{q}_i = \frac{dq_i}{dt}$$

20 Generalized Momentum

20.1 Definition

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

20.2 Examples

Cartesian coordinate:

$$L = \frac{1}{2}m\dot{x}^2 \Rightarrow p_x = m\dot{x}$$

Polar coordinate:

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$$

$$p_r = m\dot{r}, \quad p_\theta = mr^2\dot{\theta}$$

21 Cyclic Coordinates

21.1 Definition

A coordinate q_i is cyclic if:

$$\frac{\partial L}{\partial q_i} = 0$$

21.2 Result

From Euler–Lagrange equation:

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) &= 0 \\ \Rightarrow p_i &= \text{constant} \end{aligned}$$

21.3 Examples

Free particle:

$$L = \frac{1}{2} m \dot{x}^2$$

x is cyclic $\Rightarrow p_x = m\dot{x} = \text{constant}$

Central force:

$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r)$$

θ is cyclic $\Rightarrow p_\theta = mr^2 \dot{\theta} = \text{constant}$

22 Conservation Laws

22.1 Conservation of Linear Momentum

If:

$$\frac{\partial L}{\partial x} = 0$$

then:

$$p_x = \text{constant}$$

22.2 Conservation of Angular Momentum

If:

$$\frac{\partial L}{\partial \theta} = 0$$

then:

$$p_\theta = \text{constant}$$

22.3 Conservation of Energy

If the Lagrangian does not explicitly depend on time:

$$\frac{\partial L}{\partial t} = 0$$

Define energy:

$$E = \sum_i \dot{q}_i p_i - L$$

Then:

$$E = \text{constant}$$

22.4 Derivation of Energy Conservation

$$\frac{dL}{dt} = \sum_i \left(\frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \right) + \frac{\partial L}{\partial t}$$

Using Euler–Lagrange equation:

$$\frac{\partial L}{\partial q_i} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right)$$

We get:

$$\frac{d}{dt} \left(\sum_i \dot{q}_i p_i - L \right) = - \frac{\partial L}{\partial t}$$

If:

$$\frac{\partial L}{\partial t} = 0$$

then:

$$E = \sum_i \dot{q}_i p_i - L = \text{constant}$$

23 Symmetry and Conservation Laws

- Time invariance $\rightarrow$ Energy conserved
- Spatial invariance $\rightarrow$ Linear momentum conserved
- Rotational invariance $\rightarrow$ Angular momentum conserved

24 Examples

24.1 Central Force System

$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r)$$

θ is cyclic $\Rightarrow$ angular momentum conserved

24.2 Free Particle in 3D

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

x, y, z are cyclic $\Rightarrow$ momentum conserved

25 Conclusion

Lagrangian mechanics reveals deep connections between symmetry and conservation laws. Cyclic coordinates simplify problems and directly lead to conserved quantities.